AI Study Companion

Project Understanding, Scope & Prompt Reference

Prepared from our previous project discussion • September 2026

1. What I Understand the Project To Be

The AI Study Companion is intended to be a study platform, not a basic chatbot. The central idea is that students bring or connect an AI model/API key, while the platform provides the study workflow, structured features, prompts, organization, and visual experience.

The guiding product principle we established was:

“AI provides the intelligence. Our platform provides the experience.”

The platform should turn study material into useful study experiences instead of simply opening a chat screen and asking the student to talk to an AI.

2. Core Student Flow

1. Student bring study material such as notes or PDFs. (COMPUTER GENERATED PDFS)
2. The platform processes the material through the selected AI provider. (OPENROUTER, GROQ, GEMINI) (HUGGING, AZURE, LOCAL OPENSOURCE)
3. The platform generates structured outputs such as summaries, flashcards, quizzes, and note-grounded Q&A. (ALREADY PROCESSED SHOULD BE SHOWED) TO BE DISCUSSED
4. The student studies through visual, interactive interfaces rather than only a chat interface. CLEARED
5. The platform tracks study activity and progress. CLEARED
6. The student can return to the material for revision and continue studying. SESSION BASED STORADE, LAST CHECKPOINT.

3. Core Features We Discussed

Notes / PDF input: Students can provide their own study material.

Summaries: Turn uploaded material into structured summaries.

Flashcards: Generate study cards from the student's material.

Quizzes: Generate questions based on the provided study content.

Note-grounded Q&A: Allow students to ask questions grounded in their own material.

Study plans: Help organize study around the student's material and goals.

Progress tracking: Track study activity and progress over time.

Rich visualization: Use cards, progress indicators, quizzes, flashcards, timelines, diagrams (mind map/ html), and similar UI rather than making the product feel like a plain chatbot.

4. AI Provider Philosophy

The project should not be locked to one AI provider. We discussed allowing students to bring/use providers such as Gemini, OpenAI, or suitable open-source/other models. The important architectural idea is a provider-independent AI layer.

This means the platform should own the study experience and structured workflows, while the model acts as the intelligence layer.

5. Proposed Technology Direction

- Frontend: Flutter / Web
- Backend: Node.js or Laravel
- Database: PostgreSQL
- AI: Gemini, OpenAI, or other compatible AI APIs/models

These were discussed as a realistic direction rather than a claim that every component has already been finalized.

6. What We Deliberately Did NOT Want

- A simple ChatGPT-style chat application with a study-themed wrapper.
- A product completely dependent on one AI provider.
- A large feature set before validating the core study workflow.
- AI-generated content being the only visible value; the platform should provide organization and interaction.

7. Prompt / Product Direction We Used

The following are the key prompt directions and wording captured from our previous discussion. Where the exact original prompt text is not available in the retained conversation context, this section preserves the substance rather than pretending it is a verbatim transcript.

Core product prompt

Design the AI Study Companion as a platform, not a basic chatbot. Students should be able to bring "
"their own study material and use an AI provider to turn that material into structured study experiences. "
"The platform should handle the prompts, study workflows, organization, and rich visual interfaces.

Product philosophy prompt

Instead of giving students a basic chatbot, build an experience that visualizes study information through "
"cards, progress indicators, quizzes, flashcards, timelines, diagrams, and other structured components. "
"AI provides the intelligence. Our platform provides the experience.

Provider-independent AI direction

Do not design the product around only one model. The architecture and product concept should allow "
"different AI providers such as Gemini, OpenAI, or suitable open-source models. Keep the AI layer

provider-independent "
"so the study experience remains owned by the platform.

Study-material workflow prompt

Use a student's notes or PDFs as the source material. From that material, support summaries, flashcards, "
"quizzes, note-grounded questions and answers, study plans, and progress tracking. The generated content should "
"remain connected to the student's study material rather than behaving like an unrelated general chatbot.

8. Suggested Product Mental Model

Student → Study Material → AI Processing → Structured Study Experience → Progress / Revision

The platform is therefore the layer that turns raw AI capability into a repeatable student workflow.

9. MVP Boundary

The realistic first version should prove the core loop before adding expensive or complex features. The key question is whether students find value in bringing their material into the platform and using AI-generated structured study tools.

- Upload/import study material
- Generate summary
- Generate flashcards
- Generate quizzes
- Ask questions grounded in the material
- Basic study organization
- Basic progress tracking
- Provider-independent AI integration

10. Important Note About This Document

This document is a reconstruction of the project understanding and prompt directions available from our previous discussion. It does not invent missing technical decisions or claim that unrecorded prompts were verbatim. If we later have the original project brief or previous prompt document, it can be merged into this reference and marked as the authoritative version.